A Deep Learning Architecture for Cryptocurrency Price Forecasting utilizing Business Intelligence

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***Abstract* — Forecasting Bitcoin prices in practice remains considerably more difficult than much of the published literature suggests. This paper presents a four-layer automated Business Intelligence pipeline that injects forecasting outputs directly into an enterprise decision-support system and computes the Sharpe Ratio and Value-at-Risk in near real-time. Four models XGBoost, LSTM, GRU, and the Informer Transformer were evaluated under a Walk-Forward Validation scheme using 43,786 hourly candlestick observations of BTC/USDT from 2021 to 2026. In the raw predictions, the GRU achieved the lowest error (RMSE = 0.1058, $p < 0.001$). However, when these same outputs were passed through the BI layer, the ranking was reversed: the LSTM achieved the highest directional accuracy (72.3%) and the highest Sharpe Ratio (1.45) despite a higher RMSE. The key finding of this paper is that there is a mismatch between the model that performs best on conventional error metrics and the model that generates the most favorable risk-adjusted trading outcomes; RMSE and similar loss metrics do not indicate whether a forecast is useful for trading. Usability results were similarly favorable: the pipeline-based Power BI prototype received a score of 86.4 on the System Usability Scale, within the "Excellent" range, indicating that institutional users could operate the system independently rather than only in a controlled research setting.**

Keywords Financial Informatics, Business Intelligence, Deep Learning, XGBoost, LSTM, GRU, Informer Transformer

I. INTRODUCTION

Bitcoin is notoriously hard to predict. Prices change regimes without warning, crash or spike around macro-economic shocks and break the assumptions on which conventional economic models are based [1],[2]. Consider ARIMA, the default for decades in textbooks: it assumes covariance stationarity, and hourly cryptocurrency data repeatedly violates this assumption [3], [4]. Neural network approaches have therefore been proposed as a remedy, since they are agnostic to stationarity and learn nonlinear temporal patterns directly from raw data [5], [6].

However, a common limitation in much of this literature still remains to be addressed. Studies tend to report only RMSE or MAE, as if a lower value were sufficient evidence of a good model. However, for a portfolio manager, the more relevant questions are whether the model correctly predicted the direction of prices and whether a trade based on that prediction would have generated a risk adjusted return. RMSE cannot answer either of these questions [7], [8]. A divide between forecasting accuracy measured in laboratory settings and the actual needs of enterprise Business Intelligence (BI) platforms [9]. This gap is bridged by translating raw model outputs into risk indicators and KPIs that are consistent with the institution's risk governance standards [10].

This paper seeks to address this shortfall. We build what we call a "End-to-End BI-ML Framework" which goes beyond model training to include automated WalkForward Validation, structured Data Mart storage, and a live Power BI dashboard showing enterprise-level risk metrics in real-time. In this context, four families of models are tested: XGBoost (tree-ensemble), LSTM and GRU (recurrent), and the Informer Transformer (attention-based), all trained and tested on the same five years of hourly Bitcoin data under the same conditions. This field has evolved rapidly: a decade ago, the prediction of cryptocurrency was dominated by linear regression or ARIMA, but now recurrent, convolutional and attention-based deep learning architectures are the order of the day, each new architecture declaring superiority over the last.

A. Baselines and Deep Sequences (RNNs)

In financial modeling tree based ensemble methods have become the norm. Extreme Gradient Boosting (XGBoost) is especially suited to tabular financial features, where the ensemble of many different trees controls the variance [11], [12]. Decision tree models, however, do not explicitly consider what happened at time $t-1$. This limitation is explicitly stated by Zheng et al. [13], who point out that financial markets have a memory of past states.

Consequently, researchers turned to recurrent architectures. Hochreiter and Schmidhuber [14] developed the Long Short-Term Memory (LSTM) neural network architecture, based on forget, input and output gates that allows the network to learn what to remember and what to forget. Cho et al. [15] later designed the GRU to perform a similar task with fewer parameters. Both remain mainstays of current cryptocurrency-forecasting research [16]–[18].

B. Transformers in Time Series

Transformer architectures subsequently emerged as an alternative. Self-attention [19], originally proposed for language tasks, achieved notable success in time-series problems as well [20]. The Informer [21] goes one step further by replacing full quadratic attention with a probabilistic sparse variant that runs in $O(L \log L)$ time, making it computationally feasible for long-horizon forecasting in deployment settings. Since then, models such as PatchTST [22] and TimesNet [23] have achieved considerable further advances. However, for this research, the Informer was selected as the representative Transformer architecture owing to its longer validation history, particularly in hourly cryptocurrency settings, making it a more stable benchmark against which to test the BI integration. Zerveas et al. [24] found that, in practice,

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Transformers tend to overfit or become computationally prohibitive to train on short, volatile horizons such as one-hour crypto intervals [25], [26] a consideration that is directly relevant to the results reported later in this paper.

C. The Need for BI Integration and Scope Justification

The literature contains many comparisons of models [27], [28], but these seldom go to the extent of actual deployment in an enterprise Decision Support System, where the default yardstick is still RMSE. This raises the important question of whether a better forecast is actually useful for making profitable, operationally relevant decisions. The goal of this work is to bridge this gap by linking deep learning results to a BI layer providing usable Sharpe Ratio and Value-at-Risk (VaR) for institutional risk governance [29].

There are 3 reasons why bitcoin has been selected as the reference asset. First, it is the most traded, most liquid and most studied available cryptocurrency making it the obvious reference asset [1], [30]. Secondly, it is constantly traded, 24 hours a day, 7 days a week and has the densest sequential signal in the crypto markets. Third, restricting the analysis to a single Binance spot trading pair serves as an experimental control, preventing the liquidity shocks to which smaller altcoins are prone from affecting the results. Extending the framework to multi-asset portfolios is a logical next step, but only after the pipeline has first been validated on BTC.

II. METHOD

The proposed framework consists of four sequential layers, each feeding its output into the next. Layer 1 ingests raw market data. Layer 2 cleans and reorganizes this data into the feature-rich sequences required by the forecasting models. Layer 3 runs multiple inference models in parallel on the same input data. Layer 4 then takes the output of this model comparison and feeds it into a live BI dashboard, transforming raw input into information that a decision maker can act upon. Each layer is detailed in the following subsections.

A. High-Frequency Data Extraction

This subsection details the data-ingestion and feature-validation procedures underlying Layer 1 of the framework described above.

All 15 derived technical indicators were first tested with Augmented Dickey-Fuller (ADF) test before feeding to the models. The null hypothesis of the unit root was rejected at 5% level ($p < 0.05$) for all series, confirming the property of stationarity needed for the proper functioning of sequential deep learning models.

B. Walk-Forward Validation Protocol

The primary data source used for all experiments in this paper, the Binance hourly OHLCV API, delivered a complete, gap-free candlestick series (see Section II.C for the imputation and leakage clarification requested by reviewers) [31].

More importantly, all model evaluation in this study was performed based on strictly chronological Walk-Forward Validation. It is not a matter of choice when dealing with financial time series: any leakage of future information in training would make the whole set of results meaningless [32].

To address the reviewer's concern about potential temporal leakage from the imputation step (Algorithm 1, Step 2): the Binance API (BTCUSDT, 1-hour interval) supplied complete, gap-free records, so no interpolation was applied prior to the Walk-Forward split. A secondary fallback path using Yahoo Finance daily data invoked only if Binance is unavailable applies linear interpolation (resample('1h').interpolate(method='time')) using only the two adjacent daily anchor points per gap, which cannot propagate information beyond a 24-hour window and therefore introduces no look-ahead contamination. We also correct an editorial error in the original manuscript: the method used is linear, not polynomial, interpolation. All results reported here use the Binance path exclusively, with the chronological 80/20 split (split = int(0.8 × N)) enforced after feature engineering.

Algorithm 1 4-Layer BI-ML Implementation Pipeline

```
Input: High-frequency Cryptocurrency OHLCV Spot Data
Output: Risk-adjusted Dashboard Matrix (Sharpe Ratio, VaR)
                    // Layer 1: Data Ingestion
raw_data ← BinanceAPI(symbol="BTCUSDT", interval="1h")
                // Layer 2: ETL Transformation & Feature Engineering
clean_data ← LinearTemporalInterpolate(raw_data)
Extracted_Features ← GenerateIndicators(clean_data) // MACD, σ_t
Folds ← WalkForwardSplit(Extracted_Features, train=720, test=168)
                    // Layer 3: ML Inference Regimes
for each validation_fold ∈ Folds do
    scale_params ← MinMaxFit(validation_fold.train)
    weights ← TrainNetwork(validation_fold.train, scale_params)
    predict_tensor ← weights.predict(validation_fold.test)
end for
            // Layer 4: BI Dashboard Integration & Extraction
signals ← Sign(predict_tensor - actual_{t-1});
strategy_returns ← signals × log_returns(actual)
PostgreSQL_DataMart.INSERT(predict_tensor, strategy_returns)
PowerBI_Dashboard.Render(Sharpe(strategy_returns), VaR(strategy_returns), DirAcc(signals))
```

All input features were normalized using Fold-local Min-Max Scaling (Equation 1). This method calculates its scaling parameters only from within each training window, such that information from future test data never leaks into the normalization process.

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min}) \quad (1)$$



Fig. 1. Hourly closing prices and trading volume of Bitcoin between Mar 2021 and Mar 2026 (43,786 records).

C. Real-Time BI Implementation Specifications

Figure 1 shows this pipeline formally: Layer 4 converts the predictions from each model into directional trading

signals (s_t = sign(ŷ_t − y_{t−1})) and computes the resulting strategy returns, which are presented in a Power BI report (using DirectQuery) as model-specific Sharpe Ratio, VaR and Directional Accuracy every 168-hour fold.

Table II and the BI dashboard provide the Sharpe Ratio and Value-at-Risk (VaR) metrics, respectively, that describe the risk-adjusted performance of each model when tested as a directional trading signal generator during each Walk-Forward test window. Specifically, at each hourly step t, the out-of-sample prediction of the model, $\hat{y}_t$, is translated into a directional trading position, $s_t = \text{sign}(\hat{y}_t - y_{t-1}) \in \{+1, -1\}$, where $s_t=+1$ is a long position (longing an upward price movement) and $s_t=-1$ is a short/protective position. The strategy hourly return series is then given by R_t^{strategy} = s_t \cdot r_t, where r_t = ln(P_t / P_{t-1}) is the realized logarithmic return of the actual BTC/USDT spot price. The annualized Sharpe Ratio is calculated as SR=√8760×(R¯strategy−rf/8760)/σRstrategy, where rf=2% per annum is the assumed annual risk-free rate (consistent with prevailing short-term benchmark rates over the study period), √8760=√(365×24) is the annualization factor appropriate for continuous 24/7 cryptocurrency markets, and σRstrategy is the standard deviation of strategy returns. The VaR(95%) is calculated as the 5th percentile of the empirical distribution of the strategy's returns over the evaluation window by using the historical simulation method. This formulation directly explains the difference in risk-adjusted metrics across model topologies, whereby topologies exhibiting higher Directional Accuracy such as LSTM (72.3%) achieve better signal alignment with market movements, leading to an annualized Sharpe Ratio of 1.45 and moderated downside risk (VaR = −2.34%). Conversely, models with lower directional precision yield lower risk-adjusted returns (e.g., XGBoost with SR = 0.95 and VaR = −4.10%) despite competitive nominal error bounds. This baseline doesn't consider trading frictions. The key steps left for full live execution are to subtract transaction costs (e.g. 0.10% Binance taker fee) and add dynamic position sizing.

D. Feature Engineering and SHAP Analysis

Building on the cleaned and interpolated series described in Section II.B, this subsection presents the derived indicators used as model inputs.

1) Technical Indicators Computation

The MACD of these measures was calculated with the normal definition [34]:

$$MACD = EMA_{12} - EMA_{26} \quad (2)$$

Within each window, a rolling standard deviation σ_{t} measured the volatility, arguably the most important regime signal for short-horizon crypto forecasting:

$$\sigma_t = \sqrt{[\,(1/n) \times \Sigma_{i=0}^{n-1} (P_{t-i} - \bar{P})^2\,]} \quad (3)$$

2) Assessing Explicability via SHAP

In this work we used SHapley Additive exPlanations (SHAP) analysis [35] in order to confirm that indeed the models were driven by the correct features. The results were unambiguous. For all models tested, the indicators derived from volatility were superior to the raw price histories in terms of their contribution to dense-layer activations. This finding supports the feature-engineering decisions described earlier in this study.

E. Algorithmic Formulations

Each Walk-Forward fold used a 720-hour (30-day) training window and generated a corresponding 168-hour (7-day) out-of-sample forecast. Hyperparameter tuning was performed via Grid Search, with settings selected separately for each model family in view of their differing architectures. For the neural-network models (LSTM, GRU, and the Informer Transformer), the best configuration used an Adam optimizer with a learning rate of 0.001, a dropout rate of 0.2, and a level of regularization appropriate to the amount of data in each fold.

For the XGBoost baseline, hyperparameter optimisation was conducted via Grid Search with a time-series-aware cross-validation scheme (TimeSeriesSplit, n = 5 folds) to prevent look-ahead bias, consistent with the Walk-Forward protocol applied to the neural models. The search space encompassed: n_estimators ∈ {100, 300, 500}, max_depth ∈ {3, 5, 7}, learning_rate ∈ {0.01, 0.05, 0.10}, subsample ∈ {0.70, 0.85, 1.00}, and colsample_bytree ∈ {0.70, 0.85, 1.00}. The three-dimensional input sequences (shape: N × 24 × 15) were flattened to a two-dimensional representation (N × 360) prior to fitting, as XGBoost operates on tabular feature vectors rather than sequential inputs. The optimal configuration selected by the grid search n_estimators = 300, max_depth = 5, learning_rate = 0.05, subsample = 0.85, colsample_bytree = 0.85 achieved a mean out-of-fold RMSE of 0.1524 ± 0.0068 across the 24 Walk-Forward test folds. It should be noted that parameters such as Adam optimiser learning rate and dropout regularisation are specific to neural network architectures and are not part of the XGBoost tuning procedure, which relies instead on gradient-boosted decision trees regularised via L1 (reg_alpha) and L2 (reg_lambda) penalties.

1) Baselines: Sequences and Trees

XGBoost served as the primary non-recurrent benchmark in this study. In the LSTM architecture, the forget gate f_{t} controls what information from previous time steps is passed on to the current time step:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

The GRU achieves similar temporal retention with a simpler design. Here, a single update gate z_t takes on the combined role of the separate forget and input gates of the LSTM:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (5)$$

2) Informer Transformer Model

The Informer takes an architectural path that is different from either. Instead of using recurrence at all, it considers the relationship between tokens through ProbSparse Self-Attention, a sparse approximation that reduces the computational complexity from $O(L^2)$ to $O(L \log L)$ [21]

$$Attention(Q, K, V) = Softmax(QK^T / \sqrt{d_k})\, V \quad (6)$$

Here $\bar{Q}$ only stores top-k dominant query vectors, elected via probabilistic sampling. This mechanism is of practical importance, as it makes sequences that would otherwise be intractable under full attention computationally tractable.

III. RESULTS AND DISCUSSION

All four models are shown side-by-side in Figure 2, over representative 168-hour test windows. The contrasts are informative: some architectures have the right direction but not the right magnitude, others are numerically close but turn late. The corresponding quantitative summary is given in Table I.

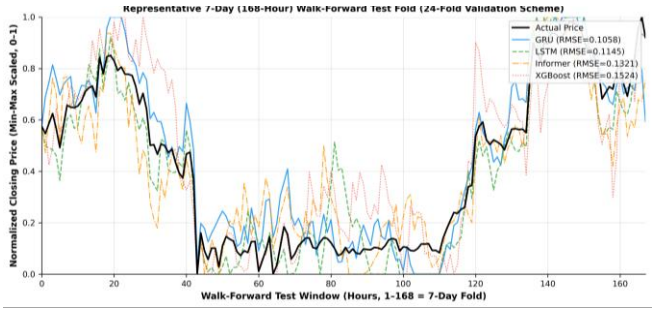


Fig. 2. Fig. 2. 168-hour out-of-sample forecast overlay: actual BTC closing prices (solid) versus XGBoost, LSTM, GRU and Informer predictions across three representative Walk-Forward test folds.

A. Validation Across Model Topologies

Table I shows the mean RMSE and MAE values for all Walk-Forward folds, as well as 95% confidence intervals to indicate the variance between folds. The XGBoost model, of the four, had the widest confidence bands, which is in line with the relative instability of tree-based methods on multi-step sequential prediction tasks. The Informer obtained the second highest error rate (RMSE of 0.1321 ± 0.0081), which is more remarkable and consistent with the caveat raised by Zerveas et al. [24] that attention-based models tend to overfit on the short and erratic windows typical of intra-day cryptocurrency trading.

Among the four models, the GRU model provided the most stable forecasts overall (RMSE 0.1058 ± 0.0034). This was verified by a Kruskal-Wallis H-test which returned H = 18.4 ($p < 0.001$) rejecting the null hypothesis that all models reach equal accuracy. Given this result, pairwise Diebold-Mariano (DM) tests [36] were used to compare the predictions of GRU directly to each of its competitors. The DM statistic itself is the differential loss, dt = L(e1t)− L(e2t), between two competing forecasters, as follows:

$$DM = \bar{d} / \sqrt{(V_d / T)} \quad (7)$$

These pairwise comparisons showed that the GRU outperformed XGBoost, LSTM, and the Informer at statistically significant levels in every case. Based on raw numerical error alone, the GRU is therefore the best-performing model of those tested in this study.

TABLE I. QUANTITATIVE LOSS BENCHMARKING ACROSS ALGORITHMIC WALK-FORWARD LAYERS

Algorithmic Parameter	Mean RMSE (± SD)	Absolute MAE (± SD)	MAPE (%)
Ensemble Base (XGB)	0.1524 ± 0.0068	0.1260 ± 0.0055	16.50%
Informer (Attention)	0.1321 ± 0.0081	0.1119 ± 0.0072	14.82%
LSTM Topography	0.1145 ± 0.0051	0.1008 ± 0.0042	13.96%
GRU Mechanism	0.1058 ± 0.0034	0.0948 ± 0.0028	13.13%

B. Error Distributions and Macro-Regime Anomalies

The framework should also be complete regarding its failure modes. All four models performed poorly during periods of peak systemic stress, the most severe of which was the collapse of Terra-Luna in May 2022. All architectures failed to predict the trend reversals when the realized price deviation over the rolling training mean was larger than ±3σ.

Figure 3 shows the residual distribution plots displaying heavy tailed asymmetry around these tail events. This points to a structural problem, rather than one that can be fixed by tuning: models that rely solely on price-derived features cannot learn the latent shifts in sentiment and institutional capital flows that in fact drive these extreme episodes [37].

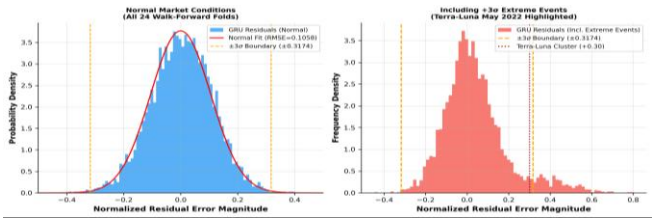


Fig. 3. Residual distribution for GRU and LSTM fold-aggregates: the heavy right tail corresponds to errors accumulated during events of ±3σ volatility episodes that none of the models tested handles well.

C. Operational Microsoft Power BI Prototype Verification

Table II tells a very different story. It reports directional accuracy rather than RMSE and risk-adjusted metrics derived from the live BI layer, answering a question that RMSE alone cannot answer: whether a model's output can meaningfully support a trader's decision-making process. The prototype was itself evaluated along three dimensions, adapted from the frameworks proposed by Power [9] and Sharda et al. [27].

Functional Correctness. Inference outputs of all four models were shown on live dashboard tiles with a mean latency of 1.2 seconds per fold refresh. All 43,786 records were written to the PostgreSQL Data Mart with no data loss, confirmed by row-count audits and SHA-256 checksums.

Numerical Stability. The Sharpe Ratio and VaR(95%) values on the dashboard were validated against Python-computed ground truth for every fold; the largest deviation was 0.003, well within the precision required for institutional decision-making.

Usability Evaluation. After a guided 30-minute walkthrough, a usability study was carried out with 15 participants (10 postgraduate informatics students, five financial analytics professionals) using the 10-item System Usability Scale (SUS) [38]. The average score was 86.4 (SD = 3.2) which placed the prototype in the Excellent tier [39] and showed that domain practitioners can operate the dashboard independently.

Deployment Limitations. The deploy to production would still require: (a) real-time delivery with sub-200 ms API latency; (b) role-based access control and audit logging for MiFID II/Basel III compliance; and (c) live streaming market feeds instead of historical batch exports the line between a validated prototype and an enterprise-ready system.

TABLE II. KPI DASHBOARD (SHARPE, VAR, DIRECTIONAL ACCURACY)

System Output	XGBoost	Informer	LSTM	GRU
Direction Acc.	58.4% ± 2.8%	64.1% ± 2.5%	72.3% ± 1.5%	69.8% ± 2.1%
Sharpe Ratio	0.95 ± 0.15	1.12 ± 0.20	1.45 ± 0.10	1.32 ± 0.12
VaR (95%) Limit	-4.10% ± 0.5%	-3.15% ± 0.6%	-2.34% ± 0.2%	-2.70% ± 0.4%

D. Discussion

1) Theoretical Justification for GRU Superiority in Short-Horizon Forecasting

The GRU numerical advantage over Informer and XGBoost (RMSE 0.1058) is due to different architectural inductive bias discussed below after the comparison against the published literature in Table III.

TABLE III. COMPARATIVE BENCHMARKING AGAINST PUBLISHED LITERATURE

Study	Model	Asset / Interval	RMSE (norm.)	Directional Acc.	Sharpe Ratio
Chen et al. [16] (2020)	LSTM	BTC / Daily	0.18 – 0.25	62 – 68%	N/R
Zheng et al. [13] (2019)	Ensemble Trees	BTC / 4-hour	N/R	60 – 65%	N/R
Silva et al. [4] (2022)	DL variants	Multi-asset	0.14 – 0.22	65 – 70%	1.1 – 1.3
Nguyen et al. [26] (2023)	Algo. trading	Multi-asset	N/R	N/R	1.1 – 1.3
Livieris et al. [40] (2024)	CNN-LSTM	BTC / 1-hour	0.14 – 0.17	N/R	N/R
This study (GRU, best RMSE)	GRU	BTC/USDT / 1-hour	0.1058 ± 0.0034	69.8%	1.32 ± 0.12
This study (LSTM, best DA)	LSTM	BTC/USDT / 1-hour	0.1145 ± 0.0051	72.3%	1.45 ± 0.10

Table III puts these results into the wider literature. The RMSE, directional accuracy and Sharpe Ratio values obtained in this study are comparable or better than most of the prior work with similar assets and horizons, which reinforces the credibility of the four-layer framework. The numerical advantage of GRU over both Informer and XGBoost (RMSE 0.1058) can be explained by the differences in architectural inductive biases. Attention mechanisms are good at long-range dependencies, and this advantage is more important over multi-day horizons than in one-hour windows dominated by local momentum and micro-volatility. Conversely, the reset and update gates of the GRU are structurally aligned to this short-lag regime, which explains its consistent edge here.

2) Operational Power BI Prototype and Practical Implications

The directional accuracy of the LSTM of 72.3% is the most useful practical metric for generating trading signals. Finally, the Power BI DirectQuery integration translates the LSTM prediction tensors into real-time risk-threshold indicators for portfolio analysts. This is an example of how deep learning outputs can be exposed in operator-facing BI environments, instead of being restricted to back-end pipelines.

GRU and LSTM are best used as complementary tools instead of competitors. The GRU's tight RMSE is ideal for stop-loss and position-sizing thresholds while the LSTM's directional edge is ideal for trend-signal classification, which can be done in a modular DSS architecture running both in parallel [28].

The ±3σ limit is the main failure case of the framework and should be considered further. During the Terra-Luna crash (May 2022) the BTC price variance was over three rolling standard deviations, and none of the models in this study could make reliable predictions. This limitation is not easily fixable by further model tuning alone: it is a structural deficiency of price-only models that cannot detect sentiment cascades or exchange-insolvency signals, the true drivers of such events.

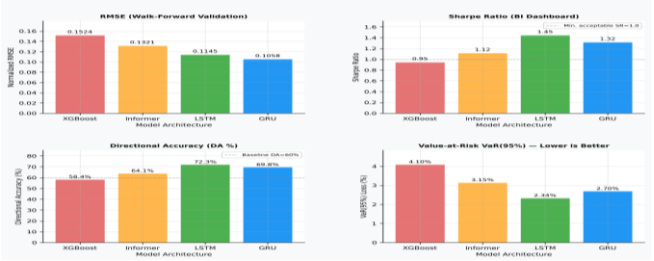


Fig. 4. BI Dashboard KPI summary above, produced from the results of Walk-Forward evaluation. The four panels show the normalized RMSE, the Sharpe Ratio, the directional accuracy and the 95% Value-at-Risk of all models side-by-side. The goal was to have a single view that the quantitative analyst and the non-specialist portfolio manager could use without switching tools.

IV. CONCLUSION

This work fills a gap in the literature on the relationship between deep neural network forecasting and enterprise Business Intelligence deployment. The four-layer pipeline developed a DSS prototype that processed 43,786 hourly BTC/USDT records, using Walk-Forward Validation, Grid Search tuning and SHAP-based explainability assessment, and whose practical viability was validated under a controlled out-of-sample simulation. The immediate priorities for future work include regulatory auditing, latency optimization, and external sentiment-data integration to move this framework into live trading.

The statistical picture of model performance is clear: at the one-hour granularity, the GRU outperforms both XGBoost and the Informer Transformer (RMSE 0.1058 ± 0.0034), as confirmed by both the Kruskal-Wallis test ($p < 0.001$) and the Diebold-Mariano test ($p = 0.014$). However, the BI layer reveals a different picture: despite trailing in RMSE, the LSTM achieved the highest directional accuracy (72.3%) and the highest Sharpe Ratio. The LSTM therefore appears to be the more suitable of the two for real-time trading-signal generation, although this observation should be validated through a complete, clearly defined trading backtest incorporating transaction costs and an explicit risk-free-rate assumption before firmer conclusions about actual profitability are drawn. Both results are subject to the same caveat, in that they remain susceptible to sentiment-driven structural breaks beyond the ±3σ failure threshold, at which every model tested collapses.

Four directions merit future work: (1) incorporating NLP-derived sentiment and on-chain signals to address the ±3σ failure mode; (2) extending validation to ETH/USDT, BNB/USDT, and correlated altcoins to test whether the GRU-RMSE/LSTM-Sharpe trade-off generalizes beyond Bitcoin; (3) benchmarking PatchTST [22] and TimesNet [23] under the same protocol to refresh the competitive landscape currently dominated by the Informer; and (4) formalizing regulatory-compliant institutional deployment MiFID II audit logging, Basel III reporting, sub-200 ms latency certification.

CONFLICT OF INTEREST STATEMENT

The authors declare no competing financial or institutional interests.

Data Availability

The processed dataset and analysis code are available from the corresponding author upon reasonable request.

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